The intervals of concavity is tabulated in Table 7.10.

Interval	$(-\infty, -1)$	$(-1, 0)$	$(0, 1)$	$(1, \infty)$
Sign of $f''(x)$	$-$	$+$	$-$	$+$
Concavity	concave down	concave up	concave down	concave up

Table 7.10

(8) As $x = -1$ and 1 are not in the domain of $f(x)$ and at $x = 0$, the second derivative is zero and $f''(x)$ changes its sign from positive to negative when passing through $x = 0$. .Therefore, the point of inflection is $(0, f(0)) = (0,0)$.

(9) $\lim\limits_{x\to\pm\infty} f(x) = \lim\limits_{x\to\pm\infty} \dfrac{3x}{x^2-1} = \lim\limits_{x\to\pm\infty} \dfrac{3}{x - \frac{1}{x}} = 0$. Therefore $y = 0$ is a horizontal asymptote. Since the denominator is zero, when $x = \pm 1$.

$\lim\limits_{x\to -1^-} \dfrac{3x}{x^2-1} = -\infty$, $\lim\limits_{x\to -1^+} \dfrac{3x}{x^2-1} = +\infty$,

$\lim\limits_{x\to 1^-} \dfrac{3x}{x^2-1} = -\infty$, $\lim\limits_{x\to 1^+} \dfrac{3x}{x^2-1} = \infty$.

Therefore $x = -1$ and $x = 1$ are vertical asymptotes.

The rough sketch of the curve is shown on the right side.

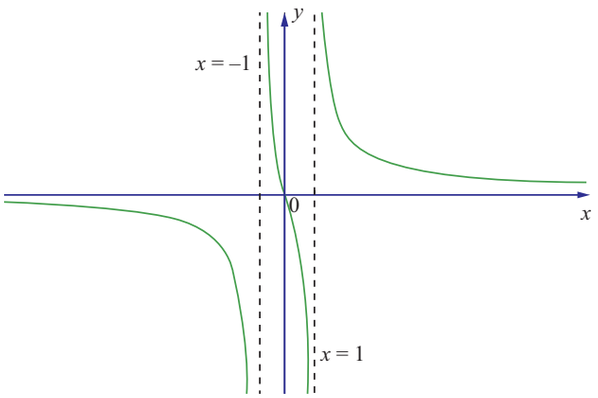


Fig.7.38

EXERCISE 7.9

1. Find the asymptotes of the following curves :

(i) $f(x) = \dfrac{x^2}{x^2-1}$ (ii) $f(x) = \dfrac{x^2}{x+1}$ (iii) $f(x) = \dfrac{3x}{\sqrt{x^2+2}}$

(iv) $f(x) = \dfrac{x^2-6x-1}{x+3}$ (v) $f(x) = \dfrac{x^2+6x-4}{3x-6}$

2. Sketch the graphs of the following functions:

(i) $y = -\dfrac{1}{3}(x^3 - 3x + 2)$ (ii) $y = x\sqrt{4-x}$ (iii) $y = \dfrac{x^2+1}{x^2-4}$

(iv) $y = \dfrac{1}{1+e^{-x}}$

Chapter 7 Differential Calculus Original-new.indd 53 26-02-2020 17:27:53